

Transformer based Prediction of Object-Object Physical Interactions for Robotic Manipulation

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Abstract

When people reach into a cluttered drawer, they instinctively predict how nudging one object will shift all the others. Robots need the same foresight: to grasp or retrieve items from clutter, they must first imagine the outcome of every push. Earlier models (Deep Interaction Prediction Network (DIPN)/Visual Foresight Trees(VFT)) predicted this by summing up pairwise object interactions, which are blind to how three or more touching objects move together. We replaced that single module with an attention-based diffusion transformer (DiT), kept every other part of the pipeline identical, and cut push-prediction error by **29.5%**, with the biggest gains in the densest clutter.



Fig.(a)UR5e Robot + ROBOTIQ gripper+ Intel Realsense D415

Background

This project builds directly on two ARC Lab projects and upgrades the one part they share.

DIPN (ICRA 2021). Given a scene and a candidate push, DIPN predicts each object's motion (Δx , Δy , $\Delta \theta$) and warps the image to "imagine" the scene after the push which then lets a planner pick pushes that separate objects for grasping. It beat prior work by 20.9% grasp success and 30.4% action efficiency.

VFT (ICRA 2022). Plugs that same predictor into a tree search (MCTS) that imagines several pushes ahead to retrieve a target. Each imagined step feeds the next, so small prediction errors compound. The predictor is the bottleneck of the whole system.

The gap. Both rely on one interaction module that sums pairwise predictions and cannot capture how 3+ touching objects move together: its error grows ~2.5x, from 0.11 with one object to 0.27 with eight. That is the metric this project aims to lower.

Objectives

- Replace the interaction module of DIPN/VFT, a sum of pairwise MLPs with multi-head self-attention over object tokens, then extend it with a diffusion transformer (DiT); segmentation, the mask encoder, and the SE(2)-and-warp decoding remain unchanged.
- Train every model from scratch on the identical ~200k-push simulation dataset and training budget, and evaluate with the same symmetric-difference metric used in the original papers.

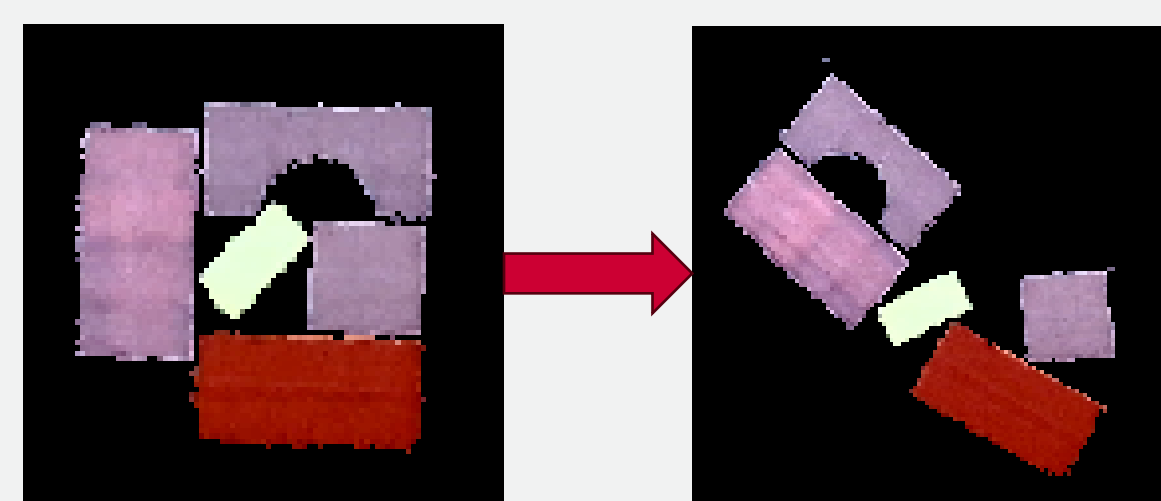
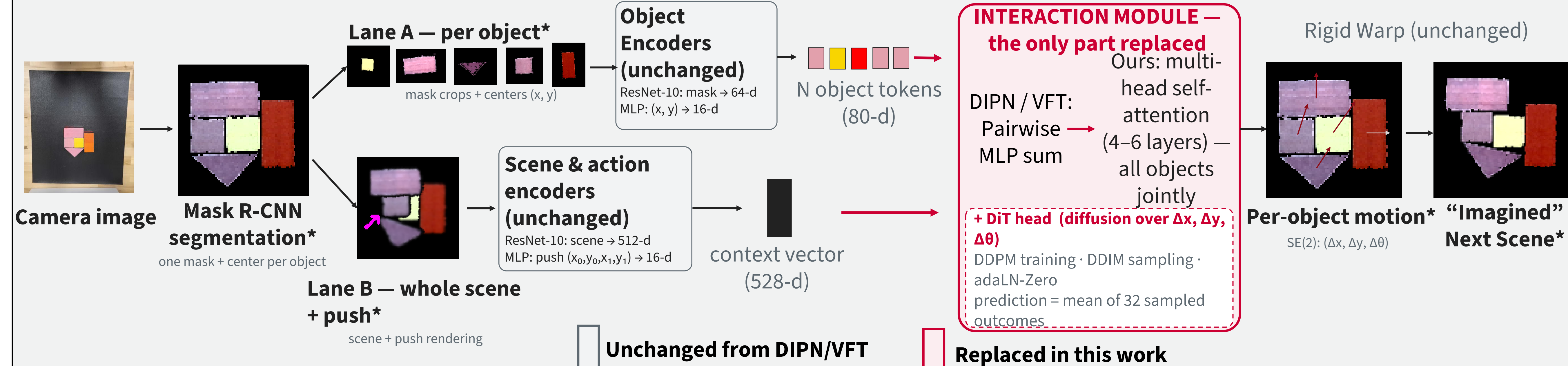


Fig.(b)

Before and after a few pushes: the target (Yellow Block) is now clear and graspable.

Methods

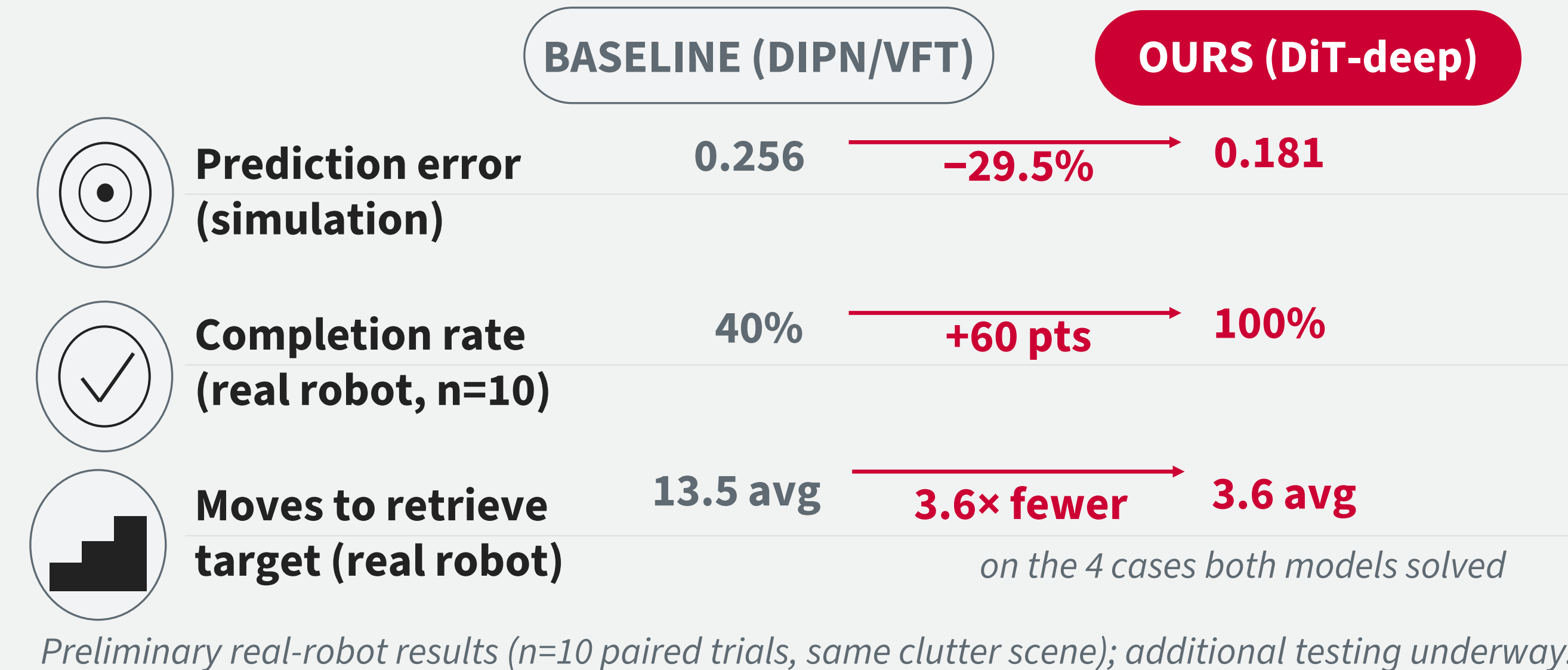


Fig(c). Flowchart depicting the stages of the model

*The images in the chart are zoomed-in versions of the actual heightmaps, segmented results used for experiments

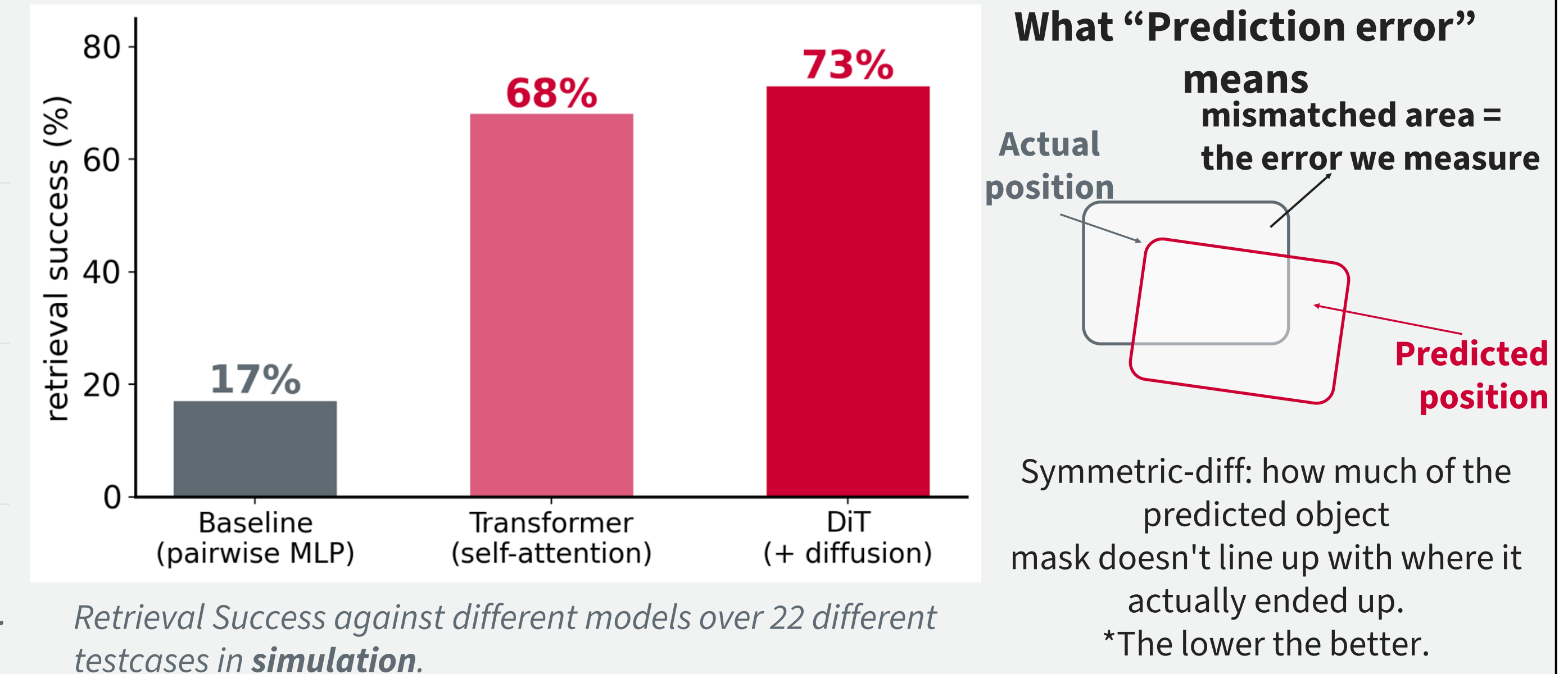
The "Imagined" Next scene is then fed to a grasp network to estimate post-push graspability, which is unchanged from the network DIPN/VFT uses to decide whether to push or grasp.

Results



Preliminary real-robot results (n=10 paired trials, same clutter scene); additional testing underway.

On 1,999 held-out simulation scenes, swapping the pairwise-MLP interaction module for self-attention alone cut prediction error 18.8%; adding a diffusion head reached 22.8%; adding capacity (DiT-deep) reached 29.5%, each step built on the last, all trained from scratch on identical data and budget. The gain concentrates in dense clutter, confirming that attention captures the multi-object interactions the original pairwise-sum architecture could not.



Retrieval Success against different models over 22 different testcases in simulation.

Across ten paired trials on the real UR5 using the same clutter scene, only the interaction module differed. DiT-deep retrieved the target in every case, while the baseline stalled and hit its 20-move search limit on six of ten, concentrated in the densest scenes. The same clutter-driven gap predicted in simulation reproduces on hardware. Results are preliminary (n=10); additional trials are underway.

Future Directions

- Expand real-robot testing** (IN PROGRESS): larger sample size for a statistically confident measurement of DiT's real-world margin
- Faster, cheaper scene understanding** (PLANNED): replace Mask R-CNN with a lighter, more precise, lower-cost alternative
- Harder test scenarios** (PLANNED): denser, adversarial clutter arrangements and object types

References/Acknowledgments

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- [1] B. Huang, S. D. Han, A. Boularias, and J. Yu, "DIPN: Deep Interaction Prediction Network with Application to Clutter Removal," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2021.
- [2] B. Huang, S. D. Han, J. Yu, and A. Boularias, "Visual Foresight Trees for Object Retrieval from Clutter with Nonprehensile Rearrangement," *IEEE Robotics and Automation Letters*, vol. 7, no. 1, pp. 231-238, 2022.